

1. Background

```
#include "library.h"
```

```
void circle_steps(double step_length, double turn_angle, int steps_left)
{
    if (steps_left == 0)
    {
        return;
    }
    note_position();
    draw_distance(step_length);
    turn_right_by_degrees(turn_angle);
    circle_steps(step_length, turn_angle, steps_left - 1);
}
```

```
void sun_or_moon(double x, double y, double radius, const int steps, int choose_sky)
{
    double pi = acos(-1.0);
    double circumference = 2 * pi * radius;
    double step_length = circumference / steps;
    double turn_angle = 360 / steps;

    start_shape();
    move_to(x + radius, y);
    set_heading_degrees(90);

    circle_steps(step_length, turn_angle, steps);
    note_position();

    if (choose_sky == 1)
        set_pen_color(color::yellow);
    else set_pen_color(color::light_grey);

    fill_shape();
}
```

```
void grass()
{
    set_pen_color(color::light_green);
}
```

```

        fill_rectangle(0,550,800,570);
    }

void dirt()
{
    set_pen_color(color::brown);
    fill_rectangle(0,570,800,600);
}

void background()
{
    make_window(800, 600);

    int choose_sky = random_in_range(1, 2);

    //background
    if (choose_sky == 1)
        set_pen_color(color::light_blue);
    else
        set_pen_color(color::dark_blue);
    fill_rectangle(0, 0, 800, 550);

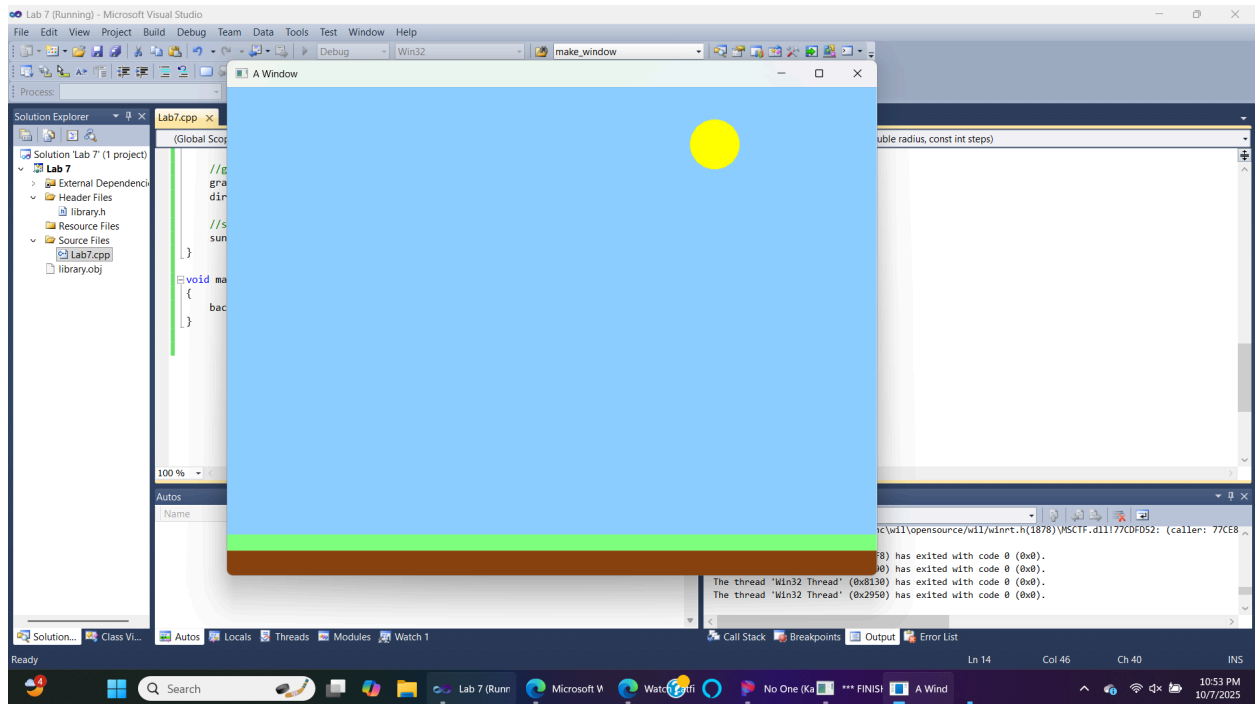
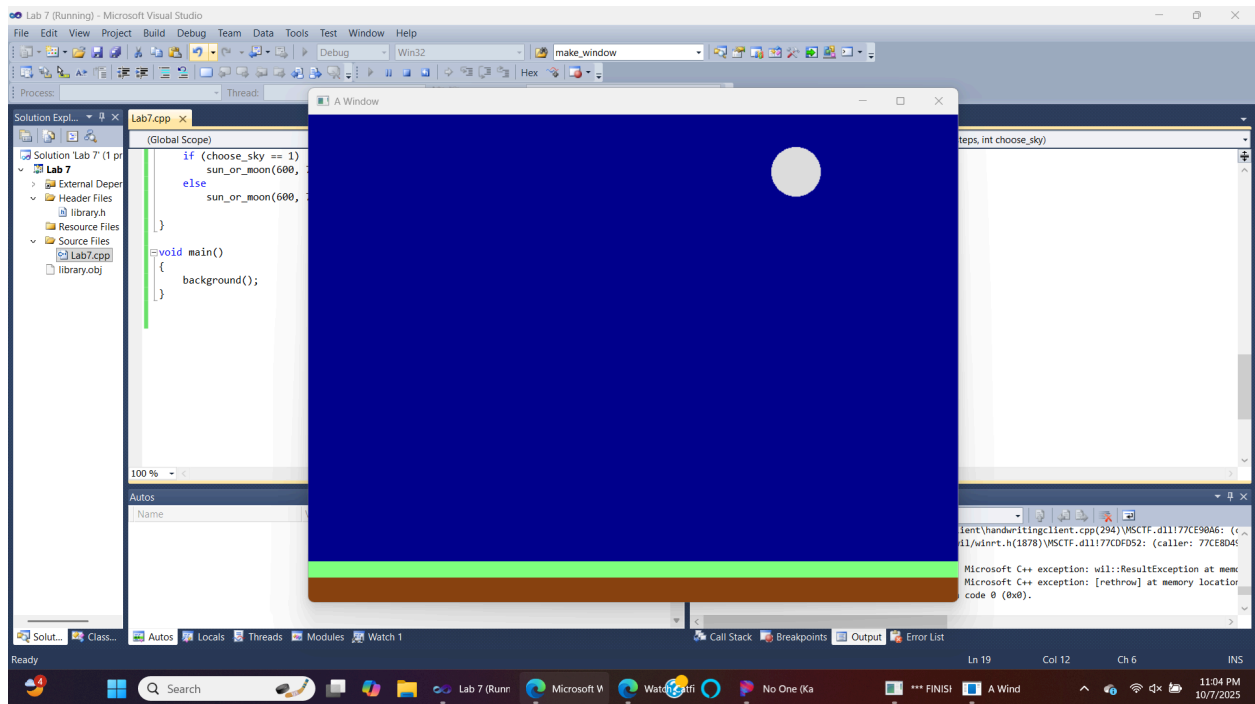
    //ground
    grass();
    dirt();

    //sun or moon or wateva
    if (choose_sky == 1)
        sun_or_moon(600, 70, 30, 360, 1);
    else
        sun_or_moon(600, 70, 30, 360, 2);

}

void main()
{
    background();
}

```



2. Office Buildings

a) *draw a single window*

```
#include "library.h"
```

```
void circle_steps(double step_length, double turn_angle, int steps_left)
{
    if (steps_left == 0)
    {
        return;
    }
    note_position();
    draw_distance(step_length);
    turn_right_by_degrees(turn_angle);
    circle_steps(step_length, turn_angle, steps_left - 1);
}
```

```
void sun_or_moon(double x, double y, double radius, const int steps, int choose_sky)
{
    double pi = acos(-1.0);
    double circumference = 2 * pi * radius;
    double step_length = circumference / steps;
    double turn_angle = 360 / steps;

    start_shape();
    move_to(x + radius, y);
    set_heading_degrees(90);

    circle_steps(step_length, turn_angle, steps);
    note_position();

    if (choose_sky == 1)
        set_pen_color(color::yellow);
    else set_pen_color(color::light_grey);

    fill_shape();
}
```

```
void grass()
```

```
{
    set_pen_color(color::light_green);
    fill_rectangle(0,550,800,570);
}
```

```
void dirt()
{
    set_pen_color(color::brown);
    fill_rectangle(0,570,800,600);
}
```

```
void background()
{
    make_window(800, 600);

    int choose_sky = random_in_range(1, 2);

    //background
    if (choose_sky == 1)
        set_pen_color(color::light_blue);
    else
        set_pen_color(color::dark_blue);
    fill_rectangle(0, 0, 800, 550);

    //ground
    grass();
    dirt();

    //sun or moon or wateva
    if (choose_sky == 1)
        sun_or_moon(600, 70, 30, 360, 1);
    else
        sun_or_moon(600, 70, 30, 360, 2);
}
```

```
void building(const int x, const int y, const int w, const int h)
{
    const int width = x + w;
    const int height = y + h;

    set_pen_color(color::pink);
```

```

        fill_rectangle(x, y, w, h);
    }

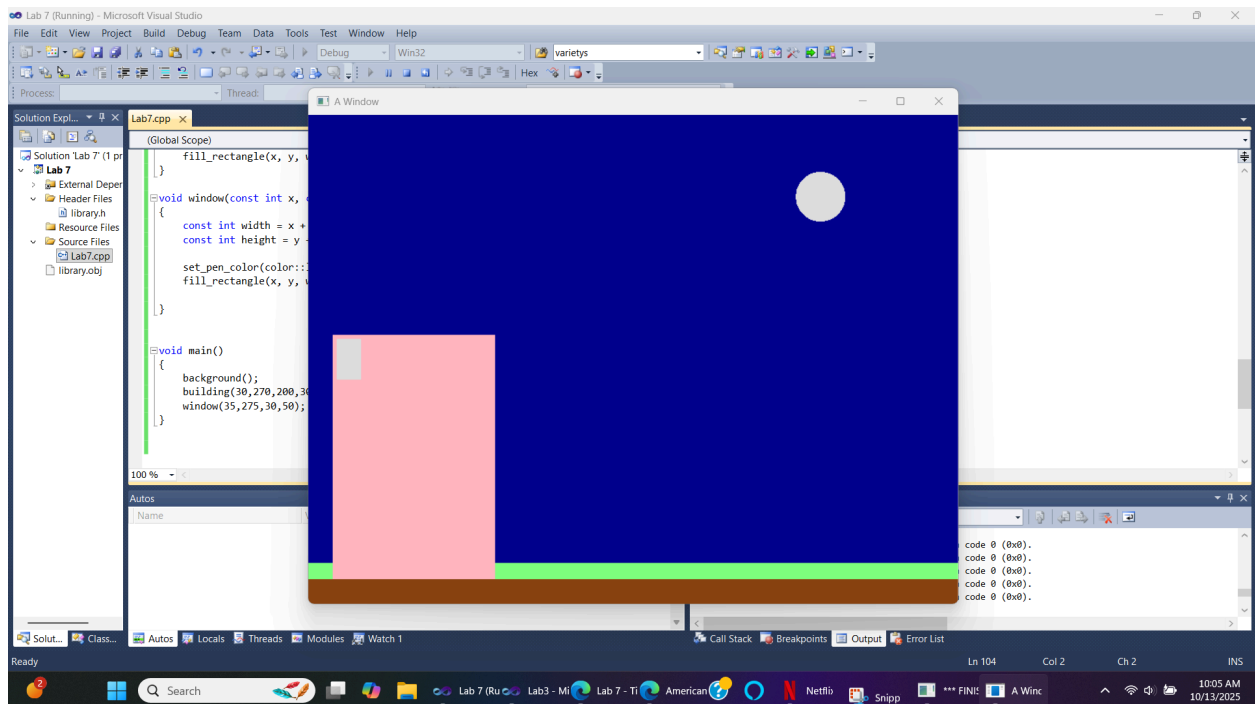
void window(const int x, const int y, const int w, const int h)
{
    const int width = x + w;
    const int height = y + h;

    set_pen_color(color::light_grey);
    fill_rectangle(x, y, w, h);

}

void main()
{
    background();
    building(30,270,200,300);
    window(35,275,30,50);
}

```



b) draw a row of windows to make a single floor

```
#include "library.h"
```

```
void circle_steps(double step_length, double turn_angle, int steps_left)
{
    if (steps_left == 0)
    {
        return;
    }
    note_position();
    draw_distance(step_length);
    turn_right_by_degrees(turn_angle);
    circle_steps(step_length, turn_angle, steps_left - 1);
}
```

```
void sun_or_moon(double x, double y, double radius, const int steps, int choose_sky)
{
    double pi = acos(-1.0);
    double circumference = 2 * pi * radius;
    double step_length = circumference / steps;
    double turn_angle = 360 / steps;

    start_shape();
    move_to(x + radius, y);
    set_heading_degrees(90);

    circle_steps(step_length, turn_angle, steps);
    note_position();

    if (choose_sky == 1)
        set_pen_color(color::yellow);
    else set_pen_color(color::light_grey);

    fill_shape();
}
```

```
void grass()
{
    set_pen_color(color::light_green);
    fill_rectangle(0,550,800,570);
}
```

```
}
```

```
void dirt()
```

```
{  
    set_pen_color(color::brown);  
    fill_rectangle(0,570,800,600);  
}
```

```
void background()
```

```
{  
    make_window(800, 600);  
  
    int choose_sky = random_in_range(1, 2);  
  
    //background  
    if (choose_sky == 1)  
        set_pen_color(color::light_blue);  
    else  
        set_pen_color(color::dark_blue);  
    fill_rectangle(0, 0, 800, 550);  
  
    //ground  
    grass();  
    dirt();  
  
    //sun or moon or wateva  
    if (choose_sky == 1)  
        sun_or_moon(600, 70, 30, 360, 1);  
    else  
        sun_or_moon(600, 70, 30, 360, 2);  
}
```

```
void building(const int x, const int y, const int w, const int h)
```

```
{  
    const int width = x + w;  
    const int height = y + h;  
  
    set_pen_color(color::pink);  
    fill_rectangle(x, y, w, h);  
}
```

```

void window(const int x, const int y, const int w, const int h)
{
    const int width = x + w;
    const int height = y + h;

    fill_rectangle(x, y, w, h);
}

```

```

void row_of_windows(const int x, const int y, const int w, const int h, const int n, const int c)
{
    int number_of_windows = 0;
    int space = 8.5;
    int current_x = x;

    if (c == 1)
        set_pen_color(color::black);
    if (c == 2)
        set_pen_color(color::orange);
    if (c == 3)
        set_pen_color(color::pink);
    if (c == 4)
        set_pen_color(color::purple);
    if (c == 5)
        set_pen_color(color::red);
    if (c == 6)
        set_pen_color(color::light_green);
    if (c == 7)
        set_pen_color(color::green);
    if (c == 8)
        set_pen_color(color::yellow);
    if (c == 9)
        set_pen_color(color::violet);
    if (c == 10)
        set_pen_color(color::brown);
    if (c == 11)
        set_pen_color(color::light_grey);
    if (c == 12)
        set_pen_color(color::white);
    if (c == 13)
        set_pen_color(color::grey);
    if (c == 14)
        set_pen_color(color::dark_blue);
}

```

```

while (number_of_windows < n)
{
    current_x = current_x + w + space;
    window(current_x, y, w, h);

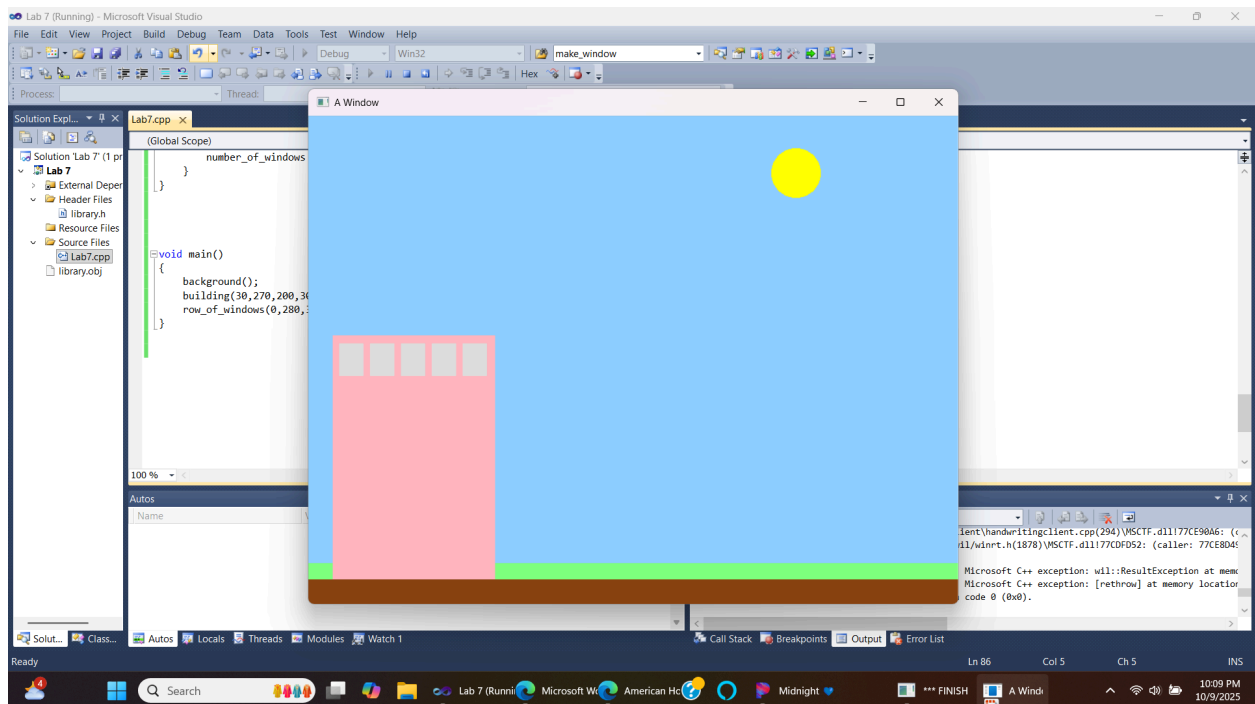
    number_of_windows = number_of_windows + 1;
}
}

```

```

void main()
{
    background();
    building(30,270,200,300);
    row_of_windows(0,280,30,40,5,11);
}

```



c) Draw a whole block of windows

```
#include "library.h"
```

```
void circle_steps(double step_length, double turn_angle, int steps_left)
{
    if (steps_left == 0)
    {
        return;
    }
    note_position();
    draw_distance(step_length);
    turn_right_by_degrees(turn_angle);
    circle_steps(step_length, turn_angle, steps_left - 1);
}
```

```
void sun_or_moon(double x, double y, double radius, const int steps, int choose_sky)
{
    double pi = acos(-1.0);
    double circumference = 2 * pi * radius;
    double step_length = circumference / steps;
    double turn_angle = 360 / steps;

    start_shape();
    move_to(x + radius, y);
    set_heading_degrees(90);

    circle_steps(step_length, turn_angle, steps);
    note_position();

    if (choose_sky == 1)
        set_pen_color(color::yellow);
    else set_pen_color(color::light_grey);

    fill_shape();
}
```

```
void grass()
{
    set_pen_color(color::light_green);
    fill_rectangle(0,550,800,570);
}
```

```
}
```

```
void dirt()
```

```
{  
    set_pen_color(color::brown);  
    fill_rectangle(0,570,800,600);  
}
```

```
void background()
```

```
{  
    make_window(800, 600);  
  
    int choose_sky = random_in_range(1, 2);  
  
    //background  
    if (choose_sky == 1)  
        set_pen_color(color::light_blue);  
    else  
        set_pen_color(color::dark_blue);  
    fill_rectangle(0, 0, 800, 550);  
  
    //ground  
    grass();  
    dirt();  
  
    //sun or moon or wateva  
    if (choose_sky == 1)  
        sun_or_moon(600, 70, 30, 360, 1);  
    else  
        sun_or_moon(600, 70, 30, 360, 2);  
}
```

```
void building(const int x, const int y, const int w, const int h)
```

```
{  
    const int width = x + w;  
    const int height = y + h;  
  
    set_pen_color(color::pink);  
    fill_rectangle(x, y, w, h);  
}
```

```

void window(const int x, const int y, const int w, const int h)
{
    const int width = x + w;
    const int height = y + h;

    fill_rectangle(x, y, w, h);
}

```

```

void row_of_windows(const int x, const int y, const int w, const int h, const int n, const int c)
{
    int number_of_windows = 0;
    double space = 8.5;
    int current_x = x;

    if (c == 1)
        set_pen_color(color::black);
    if (c == 2)
        set_pen_color(color::orange);
    if (c == 3)
        set_pen_color(color::pink);
    if (c == 4)
        set_pen_color(color::purple);
    if (c == 5)
        set_pen_color(color::red);
    if (c == 6)
        set_pen_color(color::light_green);
    if (c == 7)
        set_pen_color(color::green);
    if (c == 8)
        set_pen_color(color::yellow);
    if (c == 9)
        set_pen_color(color::violet);
    if (c == 10)
        set_pen_color(color::brown);
    if (c == 11)
        set_pen_color(color::light_grey);
    if (c == 12)
        set_pen_color(color::white);
    if (c == 13)
        set_pen_color(color::grey);
    if (c == 14)
        set_pen_color(color::dark_blue);
}

```

```

while (number_of_windows < n)
{

    current_x = current_x + w + space;
    window(current_x, y, w, h);

    number_of_windows = number_of_windows + 1;

}

```

```

void block_of_windows(const int x, const int y, const int w, const int h, const int n_floors, const
int n_windows, double space, const int c)
{
    int number_of_floors = 0;
    int s = space;
    int current_y = y;

    if (c == 1)
        set_pen_color(color::black);
    if (c == 2)
        set_pen_color(color::orange);
    if (c == 3)
        set_pen_color(color::pink);
    if (c == 4)
        set_pen_color(color::purple);
    if (c == 5)
        set_pen_color(color::red);
    if (c == 6)
        set_pen_color(color::light_green);
    if (c == 7)
        set_pen_color(color::green);
    if (c == 8)
        set_pen_color(color::yellow);
    if (c == 9)
        set_pen_color(color::violet);
    if (c == 10)
        set_pen_color(color::brown);
    if (c == 11)
        set_pen_color(color::light_grey);
    if (c == 12)

```

```

        set_pen_color(color::white);
    if (c == 13)
        set_pen_color(color::grey);
    if (c == 14)
        set_pen_color(color::dark_blue);

    while (number_of_floors < n_floors)
    {

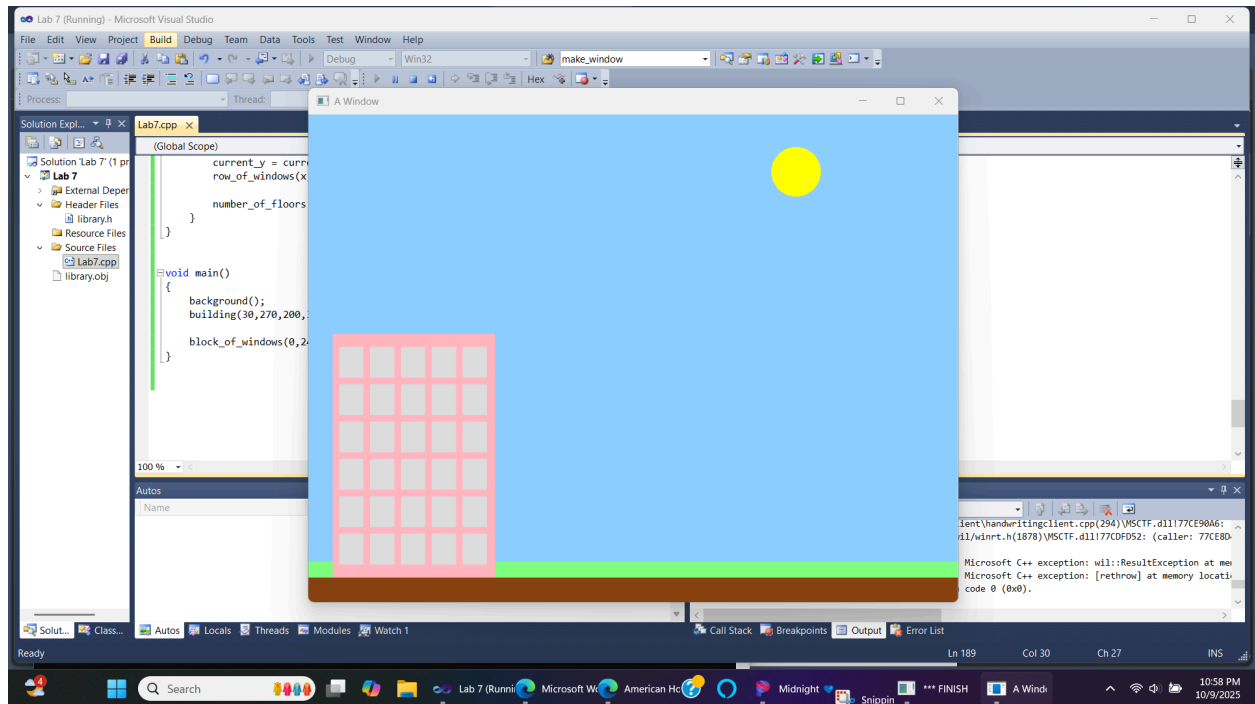
        current_y = current_y + h + space;
        row_of_windows(x, current_y, w, h, n_windows, c);

        number_of_floors = number_of_floors + 1;
    }
}

void main()
{
    background();
    building(30,270,200,300);

    block_of_windows(0,240,30,38,6,5,8.5,11);
}

```



d) Draw a whole office building

```
#include "library.h"
```

```
void circle_steps(double step_length, double turn_angle, int steps_left)
{
    if (steps_left == 0)
    {
        return;
    }
    note_position();
    draw_distance(step_length);
    turn_right_by_degrees(turn_angle);
    circle_steps(step_length, turn_angle, steps_left - 1);
}
```

```
void sun_or_moon(double x, double y, double radius, const int steps, int choose_sky)
{
    double pi = acos(-1.0);
    double circumference = 2 * pi * radius;
    double step_length = circumference / steps;
    double turn_angle = 360 / steps;
```

```

        start_shape();
        move_to(x + radius, y);
        set_heading_degrees(90);

        circle_steps(step_length, turn_angle, steps);
        note_position();

        if (choose_sky == 1)
            set_pen_color(color::yellow);
        else set_pen_color(color::light_grey);

        fill_shape();
    }

void grass()
{
    set_pen_color(color::light_green);
    fill_rectangle(0,550,800,570);
}

void dirt()
{
    set_pen_color(color::brown);
    fill_rectangle(0,570,800,600);
}

void background()
{
    make_window(800, 600);

    int choose_sky = random_in_range(1, 2);

    //background
    if (choose_sky == 1)
        set_pen_color(color::light_blue);
    else
        set_pen_color(color::dark_blue);
    fill_rectangle(0, 0, 800, 550);

    //ground

```

```

    grass();
    dirt();

    //sun or moon or wateva
    if (choose_sky == 1)
        sun_or_moon(600, 70, 30, 360, 1);
    else
        sun_or_moon(600, 70, 30, 360, 2);
}

void building(const int x, const int y, const int w, const int h)
{
    const int width = x + w;
    const int height = y + h;

    set_pen_color(color::pink);
    fill_rectangle(x, y, w, h);
}

void window(const int x, const int y, const int w, const int h)
{
    const int width = x + w;
    const int height = y + h;

    fill_rectangle(x, y, w, h);
}

void row_of_windows(const int x, const int y, const int w, const int h, const int n, const int c)
{
    int number_of_windows = 0;
    double space = 8.5;
    int current_x = x;

    if (c == 1)
        set_pen_color(color::orange);
    if (c == 2)
        set_pen_color(color::pink);
    if (c == 3)
        set_pen_color(color::purple);
    if (c == 4)
        set_pen_color(color::light_red);
}

```

```

        if (c == 5)
            set_pen_color(color::indigo);
        if (c == 6)
            set_pen_color(color::light_green);
        if (c == 7)
            set_pen_color(color::green);
        if (c == 8)
            set_pen_color(color::yellow);
        if (c == 9)
            set_pen_color(color::violet);
        if (c == 10)
            set_pen_color(color::red);
        if (c == 11)
            set_pen_color(color::light_grey);
        if (c == 12)
            set_pen_color(color::white);
        if (c == 13)
            set_pen_color(color::grey);
        if (c == 14)
            set_pen_color(color::dark_blue);

        while (number_of_windows < n)
        {

            current_x = current_x + w + space;
            window(current_x, y, w, h);

            number_of_windows = number_of_windows + 1;
        }
    }

void block_of_windows(const int x, const int y, const int w, const int h, const int n_floors, const
int n_windows, double space, const int c)
{
    int number_of_floors = 0;
    int s = space;
    int current_y = y;

    if (c == 1)
        set_pen_color(color::orange);
    if (c == 2)

```

```

        set_pen_color(color::pink);
    if (c == 3)
        set_pen_color(color::purple);
    if (c == 4)
        set_pen_color(color::light_red);
    if (c == 5)
        set_pen_color(color::indigo);
    if (c == 6)
        set_pen_color(color::light_green);
    if (c == 7)
        set_pen_color(color::green);
    if (c == 8)
        set_pen_color(color::yellow);
    if (c == 9)
        set_pen_color(color::violet);
    if (c == 10)
        set_pen_color(color::red);
    if (c == 11)
        set_pen_color(color::light_grey);
    if (c == 12)
        set_pen_color(color::white);
    if (c == 13)
        set_pen_color(color::grey);
    if (c == 14)
        set_pen_color(color::dark_blue);

    while (number_of_floors < n_floors)
    {
        row_of_windows(x, current_y, w, h, n_windows, c);
        current_y = current_y + h + space;

        number_of_floors = number_of_floors + 1;
    }
}

void office_building(const int x, const int ground_y, int w)
{
    const int n_floors = random_in_range(4, 10);
    int window_width = 30;
    int window_height = 35;

```

```

int space = 8;
int top_bottom_margin = 10;

int building_height = n_floors * (window_height + space) + top_bottom_margin * 2;

//STAY ON THE GROUND!!!!
int y = 570 - building_height;

const int building_color = random_in_range(1, 14);
fill_rectangle(x, y, w, building_height);
int window_color = random_in_range(1, 14);

while (window_color == building_color)
{
    window_color = random_in_range(1, 14);
}

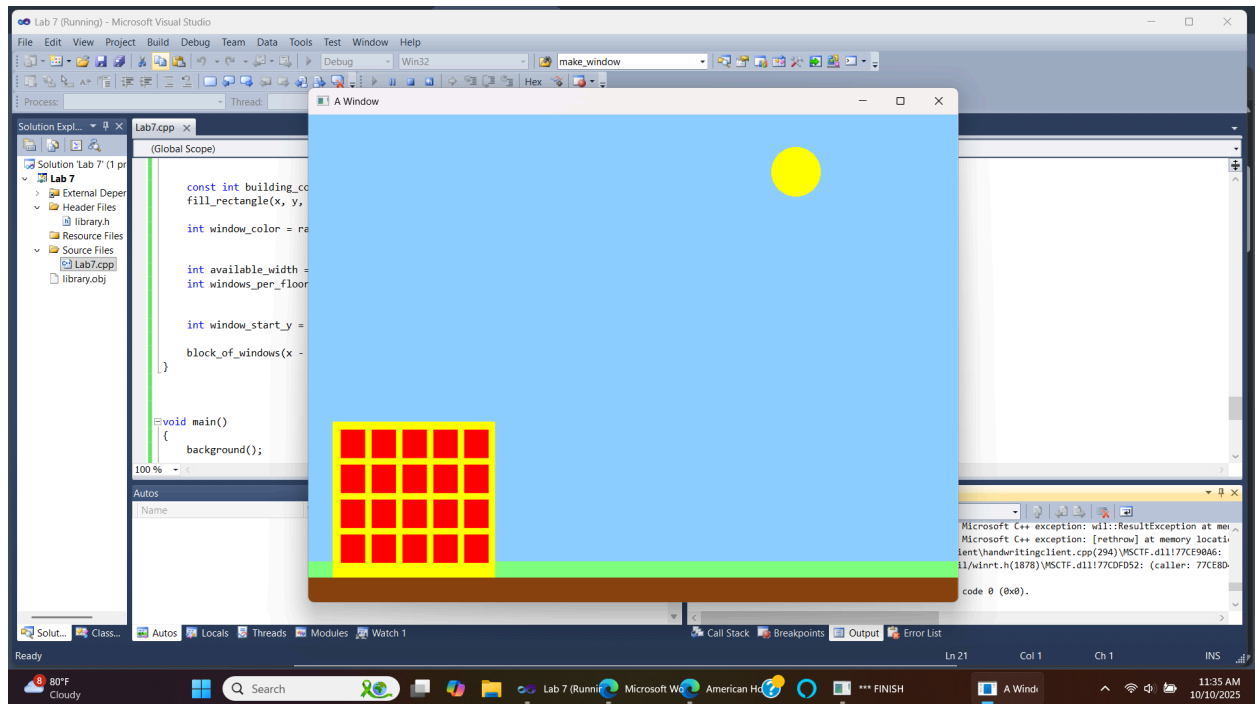
int available_width = w - space;
int windows_per_floor = available_width / (window_width + space);

int window_start_y = y + top_bottom_margin;

    block_of_windows(x - 27.5, window_start_y, window_width, window_height, n_floors,
windows_per_floor, space, window_color);
}

void main()
{
    background();
    office_building(30, 180, 200);
}

```



e) Draw a city scene

```
#include "library.h"
```

```
void circle_steps(double step_length, double turn_angle, int steps_left)
{
    if (steps_left == 0)
    {
        return;
    }
    note_position();
    draw_distance(step_length);
    turn_right_by_degrees(turn_angle);
    circle_steps(step_length, turn_angle, steps_left - 1);
}
```

```
void sun_or_moon(double x, double y, double radius, const int steps, int choose_sky)
{
    double pi = acos(-1.0);
    double circumference = 2 * pi * radius;
    double step_length = circumference / steps;
    double turn_angle = 360 / steps;
```

```

        start_shape();
        move_to(x + radius, y);
        set_heading_degrees(90);

        circle_steps(step_length, turn_angle, steps);
        note_position();

        if (choose_sky == 1)
            set_pen_color(color::yellow);
        else set_pen_color(color::light_grey);

        fill_shape();
    }

void grass()
{
    set_pen_color(color::light_green);
    fill_rectangle(0,550,800,570);
}

void dirt()
{
    set_pen_color(color::brown);
    fill_rectangle(0,570,800,600);
}

void background()
{
    make_window(800, 600);

    int choose_sky = random_in_range(1, 2);

    //background
    if (choose_sky == 1)
        set_pen_color(color::light_blue);
    else
        set_pen_color(color::dark_blue);
    fill_rectangle(0, 0, 800, 550);
}

```

```

    //ground
    grass();
    dirt();

    //sun or moon or wateva
    if (choose_sky == 1)
        sun_or_moon(600, 70, 30, 360, 1);
    else
        sun_or_moon(600, 70, 30, 360, 2);
}

void building(const int x, const int y, const int w, const int h)
{
    const int width = x + w;
    const int height = y + h;

    set_pen_color(color::pink);
    fill_rectangle(x, y, w, h);
}

void window(const int x, const int y, const int w, const int h)
{
    const int width = x + w;
    const int height = y + h;

    fill_rectangle(x, y, w, h);
}

void row_of_windows(const int x, const int y, const int w, const int h, const int n, const int c)
{
    int number_of_windows = 0;
    double space = 8.5;
    int current_x = x;

    if (c == 1)
        set_pen_color(color::orange);
    if (c == 2)
        set_pen_color(color::pink);
    if (c == 3)
        set_pen_color(color::purple);
    if (c == 4)

```

```

        set_pen_color(color::light_red);
    if (c == 5)
        set_pen_color(color::indigo);
    if (c == 6)
        set_pen_color(color::light_green);
    if (c == 7)
        set_pen_color(color::green);
    if (c == 8)
        set_pen_color(color::yellow);
    if (c == 9)
        set_pen_color(color::violet);
    if (c == 10)
        set_pen_color(color::red);
    if (c == 11)
        set_pen_color(color::light_grey);
    if (c == 12)
        set_pen_color(color::white);
    if (c == 13)
        set_pen_color(color::grey);
    if (c == 14)
        set_pen_color(color::dark_blue);

    while (number_of_windows < n)
    {

        current_x = current_x + w + space;
        window(current_x, y, w, h);

        number_of_windows = number_of_windows + 1;
    }
}

```

```

void block_of_windows(const int x, const int y, const int w, const int h, const int n_floors, const
int n_windows, double space, const int c)
{
    int number_of_floors = 0;
    int s = space;
    int current_y = y;

    if (c == 1)
        set_pen_color(color::orange);

```

```

    if (c == 2)
        set_pen_color(color::pink);
    if (c == 3)
        set_pen_color(color::purple);
    if (c == 4)
        set_pen_color(color::light_red);
    if (c == 5)
        set_pen_color(color::indigo);
    if (c == 6)
        set_pen_color(color::light_green);
    if (c == 7)
        set_pen_color(color::green);
    if (c == 8)
        set_pen_color(color::yellow);
    if (c == 9)
        set_pen_color(color::violet);
    if (c == 10)
        set_pen_color(color::red);
    if (c == 11)
        set_pen_color(color::light_grey);
    if (c == 12)
        set_pen_color(color::white);
    if (c == 13)
        set_pen_color(color::grey);
    if (c == 14)
        set_pen_color(color::dark_blue);

    while (number_of_floors < n_floors)
    {
        row_of_windows(x, current_y, w, h, n_windows, c);
        current_y = current_y + h + space;

        number_of_floors = number_of_floors + 1;
    }
}

void office_building(const int x, const int ground_y, int w)
{
    const int n_floors = random_in_range(4, 10);
    int window_width = 30;

```

```

int window_height = 35;
int space = 8;
int top_bottom_margin = 10;

int building_height = n_floors * (window_height + space) + top_bottom_margin * 2;

//STAY ON THE GROUND!!!!
int y = 570 - building_height;

const int building_color = random_in_range(1, 14);
fill_rectangle(x, y, w, building_height);
int window_color = random_in_range(1, 14);

while (window_color == building_color)
{
    window_color = random_in_range(1, 14);
}

int available_width = w - space;
int windows_per_floor = available_width / (window_width + space);

int window_start_y = y + top_bottom_margin;

    block_of_windows(x - 27.5, window_start_y, window_width, window_height, n_floors,
windows_per_floor, space, window_color);
}

void city_scene()
{
    int x = 20;
    int ground_y = 240;

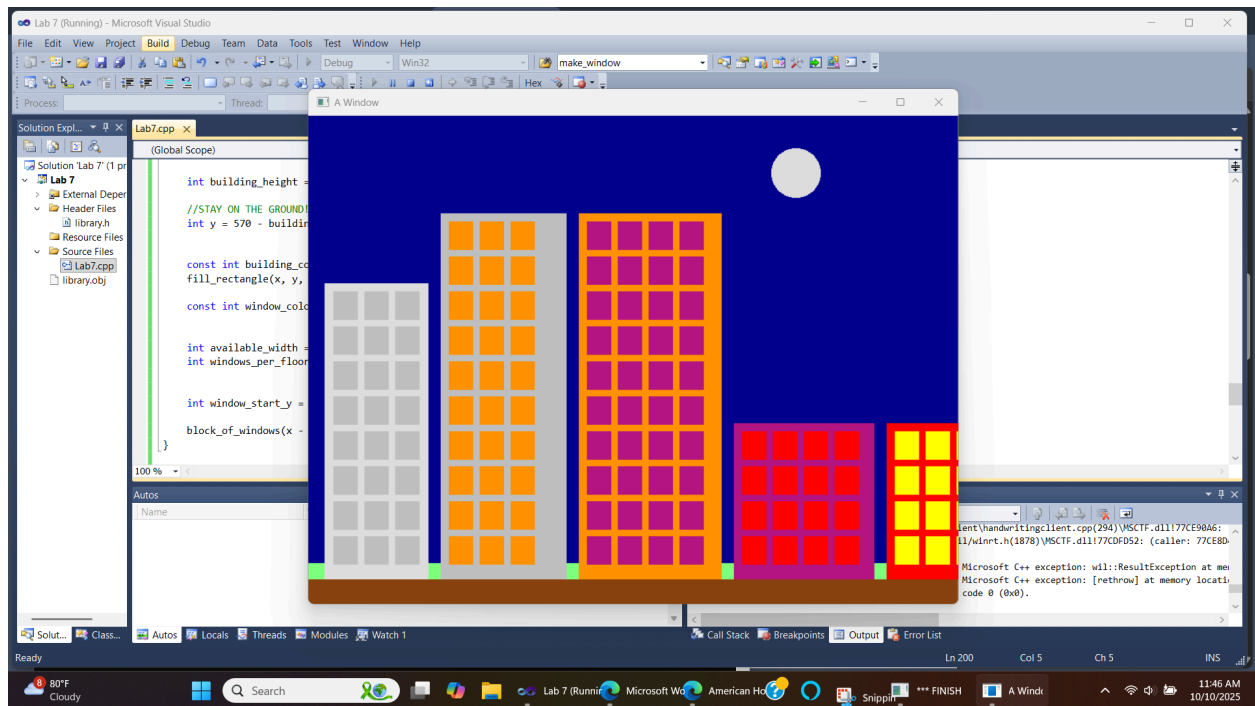
    while (x < 800)
    {
        int width = random_in_range(120, 180);
        office_building(x, ground_y, width);
        x = x + width + 15;
    }
}

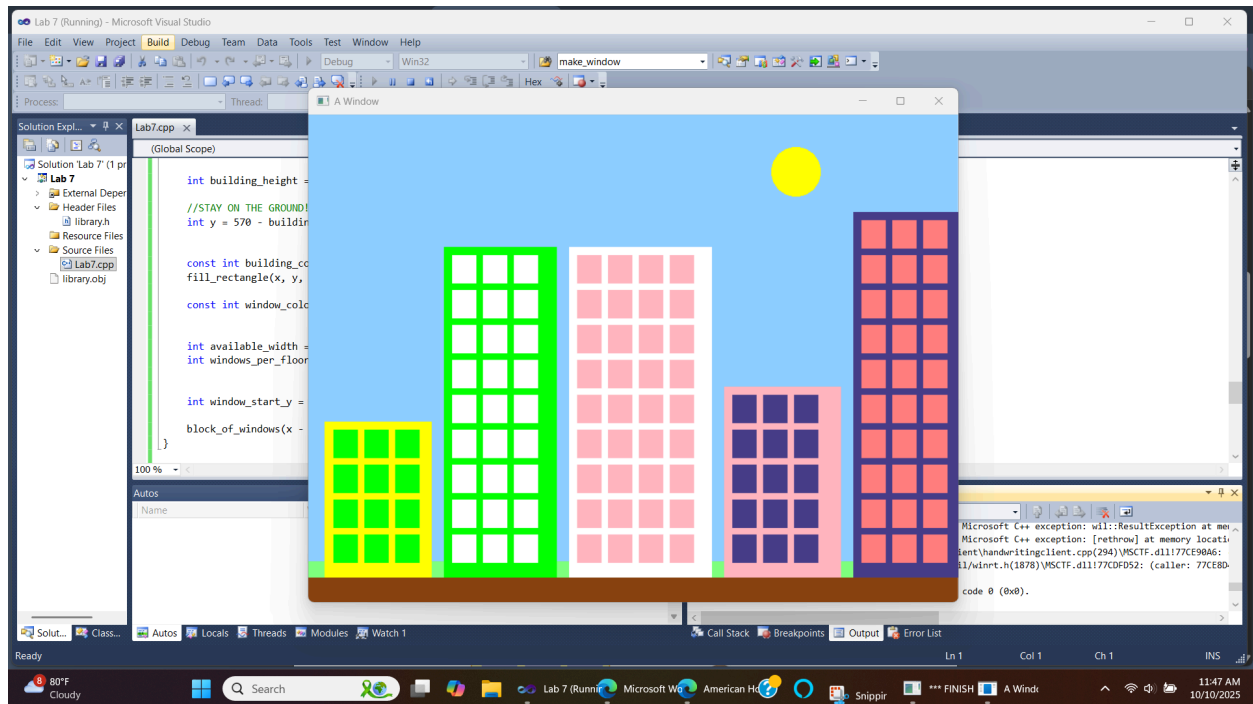
```

```

void main()
{
    background();
    city_scene();
}

```





3. Houses

```
#include "library.h"
```

```
void circle_steps(double step_length, double turn_angle, int steps_left)
{
    if (steps_left == 0)
    {
        return;
    }
    note_position();
    draw_distance(step_length);
    turn_right_by_degrees(turn_angle);
    circle_steps(step_length, turn_angle, steps_left - 1);
}
```

```
void sun_or_moon(double x, double y, double radius, const int steps, int choose_sky)
{
    double pi = acos(-1.0);
    double circumference = 2 * pi * radius;
    double step_length = circumference / steps;
```

```

double turn_angle = 360 / steps;

    start_shape();
    move_to(x + radius, y);
    set_heading_degrees(90);

    circle_steps(step_length, turn_angle, steps);
    note_position();

    if (choose_sky == 1)
        set_pen_color(color::yellow);
    else set_pen_color(color::light_grey);

    fill_shape();

}

void grass()
{
    set_pen_color(color::light_green);
    fill_rectangle(0,550,800,570);
}

void dirt()
{
    set_pen_color(color::brown);
    fill_rectangle(0,570,800,600);
}

void background()
{
    make_window(800, 600);

    int choose_sky = random_in_range(1, 2);

    //background
    if (choose_sky == 1)
        set_pen_color(color::light_blue);
    else
        set_pen_color(color::dark_blue);
    fill_rectangle(0, 0, 800, 550);
}

```

```

    //ground
    grass();
    dirt();

    //sun or moon or wateva
    if (choose_sky == 1)
        sun_or_moon(600, 70, 30, 360, 1);
    else
        sun_or_moon(600, 70, 30, 360, 2);
}

void building(const int x, const int y, const int w, const int h)
{
    const int width = x + w;
    const int height = y + h;

    set_pen_color(color::pink);
    fill_rectangle(x, y, w, h);
}

void window(const int x, const int y, const int w, const int h)
{
    const int width = x + w;
    const int height = y + h;

    fill_rectangle(x, y, w, h);
}

void row_of_windows(const int x, const int y, const int w, const int h, const int n, double
space_x, const int c)
{
    int number_of_windows = 0;
    int current_x = x;

    if (c == 1)
        set_pen_color(color::orange);
    if (c == 2)
        set_pen_color(color::pink);
    if (c == 3)
        set_pen_color(color::purple);

```

```

        if (c == 4)
            set_pen_color(color::light_red);
        if (c == 5)
            set_pen_color(color::indigo);
        if (c == 6)
            set_pen_color(color::light_green);
        if (c == 7)
            set_pen_color(color::green);
        if (c == 8)
            set_pen_color(color::yellow);
        if (c == 9)
            set_pen_color(color::violet);
        if (c == 10)
            set_pen_color(color::red);
        if (c == 11)
            set_pen_color(color::light_grey);
        if (c == 12)
            set_pen_color(color::white);
        if (c == 13)
            set_pen_color(color::grey);
        if (c == 14)
            set_pen_color(color::dark_blue);

        while (number_of_windows < n)
        {

            current_x = current_x + w + space_x;
            window(current_x, y, w, h);

            number_of_windows = number_of_windows + 1;

        }
    }

void block_of_windows(const int x, const int y, const int w, const int h, const int n_floors, const
int n_windows, double space_x, double space_y, const int c)
{
    int number_of_floors = 0;
    int sx = space_x;
    int sy = space_y;
    int current_y = y;

```

```

    if (c == 1)
        set_pen_color(color::orange);
    if (c == 2)
        set_pen_color(color::pink);
    if (c == 3)
        set_pen_color(color::purple);
    if (c == 4)
        set_pen_color(color::light_red);
    if (c == 5)
        set_pen_color(color::indigo);
    if (c == 6)
        set_pen_color(color::light_green);
    if (c == 7)
        set_pen_color(color::green);
    if (c == 8)
        set_pen_color(color::yellow);
    if (c == 9)
        set_pen_color(color::violet);
    if (c == 10)
        set_pen_color(color::red);
    if (c == 11)
        set_pen_color(color::light_grey);
    if (c == 12)
        set_pen_color(color::white);
    if (c == 13)
        set_pen_color(color::grey);
    if (c == 14)
        set_pen_color(color::dark_blue);

    while (number_of_floors < n_floors)
    {
        row_of_windows(x, current_y, w, h, n_windows, space_x, c);
        current_y = current_y + h + space_y;

        number_of_floors = number_of_floors + 1;
    }
}

void office_building(const int x, const int ground_y, int w)
{

```

```
const int n_floors = random_in_range(4, 10);
int window_width = 30;
int window_height = 35;
int space_x = 8;
int space_y = 8;
int top_bottom_margin = 10;
```

```
int building_height = n_floors * (window_height + space_y) + top_bottom_margin * 2;
```

```
//STAY ON THE GROUND!!!!
int y = 570 - building_height;
```

```
int building_color = random_in_range(1, 14);
fill_rectangle(x, y, w, building_height);
int window_color = random_in_range(1, 14);
```

```
while (window_color == building_color)
{
    window_color = random_in_range(1, 14);
}
```

```
int available_width = w - space_y;
int windows_per_floor = available_width / (window_width + space_y);
```

```
int window_start_y = y + top_bottom_margin;
```

```
block_of_windows(x - 27.5, window_start_y, window_width, window_height, n_floors,
windows_per_floor, space_x, space_y, window_color);
}
```

```
void house(const int x, const int ground_y, int w)
{
```

```
    const int n_floors = random_in_range(1, 2);
```

```
    int window_width = 30;
    int window_height = 35;
        int space_x = 30;
    int space_y = 8;
```

```

int top_bottom_margin = 10;

int house_height = n_floors * (window_height + space_y) + top_bottom_margin * 2;

int y = ground_y - house_height;

int house_color = random_in_range(1, 8);
fill_rectangle(x, y, w, house_height);

start_shape();
move_to(x, y);
note_position();
move_to(x + w / 2, y - 40);
note_position();
move_to(x + w, y);
note_position();
set_pen_color(color::brown);
fill_shape();

int available_width = w - space_y;
int windows_per_floor = available_width / (window_width + space_y);
int window_color = random_in_range(1, 14);

    while (window_color == house_color)
    {
        window_color = random_in_range(1, 14);
    }

int window_start_y = y + top_bottom_margin;
block_of_windows(x-55, window_start_y, window_width, window_height, n_floors,
windows_per_floor, space_x, space_y, window_color);

int door_color = random_in_range(1,14);
int door_x = x + 35;
int door_y = (ground_y - window_height - 5);
fill_rectangle(door_x, door_y, window_width, window_height + 10);
}

```

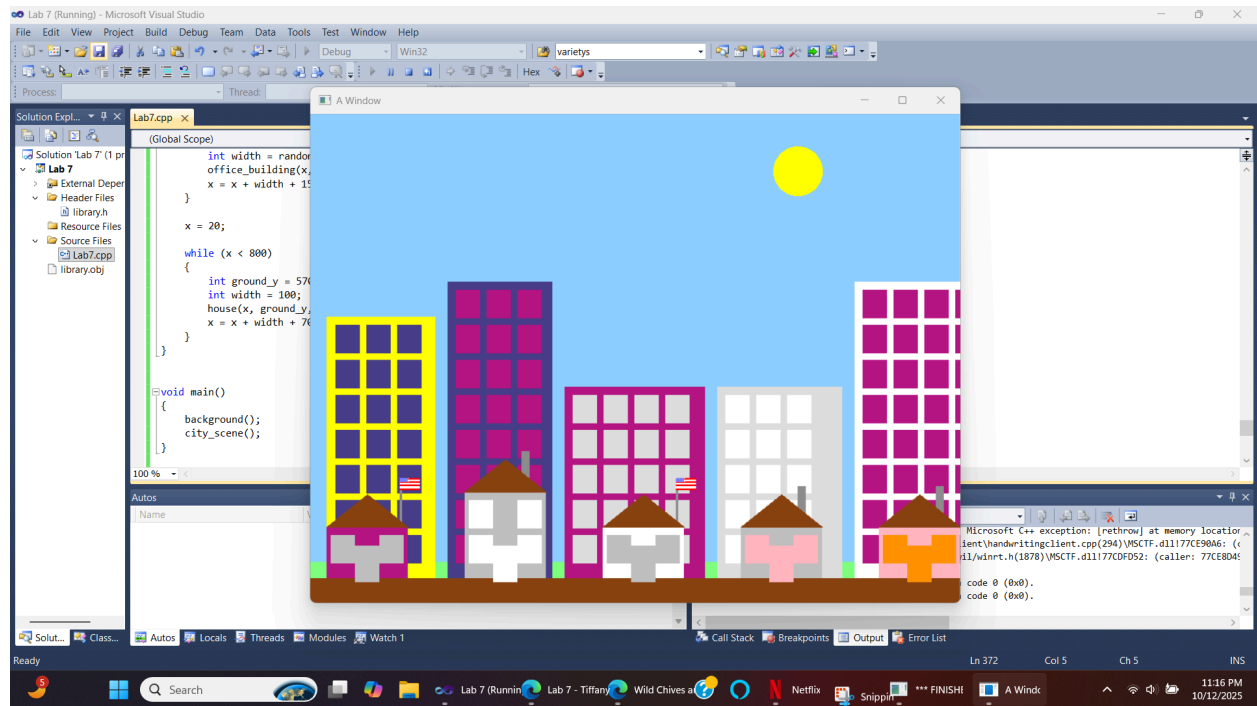
```
void city_scene()
{
    int x = 20;

    while (x < 800)
    {
        int ground_y = 240;
        int width = random_in_range(120, 180);
        office_building(x, ground_y, width);
        x = x + width + 15;
    }

    x = 20;

    while (x < 800)
    {
        int ground_y = 570;
        int width = 100;
        house(x, ground_y, width);
        x = x + width + 70;
    }
}

void main()
{
    background();
    city_scene();
}
```



4. Variety

#include "library.h"

```

void circle_steps(double step_length, double turn_angle, int steps_left)
{
    if (steps_left == 0)
    {
        return;
    }
    note_position();
    draw_distance(step_length);
    turn_right_by_degrees(turn_angle);
    circle_steps(step_length, turn_angle, steps_left - 1);
}

```

```

void sun_or_moon(double x, double y, double radius, const int steps, int choose_sky)
{
    double pi = acos(-1.0);
    double circumference = 2 * pi * radius;
    double step_length = circumference / steps;
    double turn_angle = 360 / steps;

    start_shape();
    move_to(x + radius, y);
    set_heading_degrees(90);

    circle_steps(step_length, turn_angle, steps);
    note_position();

    if (choose_sky == 1)
        set_pen_color(color::yellow);
    else set_pen_color(color::light_grey);

    fill_shape();
}

```

```

void grass()
{
    set_pen_color(color::light_green);
    fill_rectangle(0,550,800,570);
}

```

```
void dirt()
{
    set_pen_color(color::brown);
    fill_rectangle(0,570,800,600);
}
```

```
void background()
{
    make_window(800, 600);

    int choose_sky = random_in_range(1, 2);

    //background
    if (choose_sky == 1)
        set_pen_color(color::light_blue);
    else
        set_pen_color(color::dark_blue);
    fill_rectangle(0, 0, 800, 550);

    //ground
    grass();
    dirt();

    //sun or moon or wateva
    if (choose_sky == 1)
        sun_or_moon(600, 70, 30, 360, 1);
    else
        sun_or_moon(600, 70, 30, 360, 2);
}
```

```
void building(const int x, const int y, const int w, const int h)
{
    const int width = x + w;
    const int height = y + h;

    set_pen_color(color::pink);
    fill_rectangle(x, y, w, h);
}
```

```
void window(const int x, const int y, const int w, const int h)
{
    const int width = x + w;
```

```

    const int height = y + h;

    fill_rectangle(x, y, w, h);

}

void row_of_windows(const int x, const int y, const int w, const int h, const int n, double
space_x, const int c)
{
    int number_of_windows = 0;
    int current_x = x;

    if (c == 1)
        set_pen_color(color::orange);
    if (c == 2)
        set_pen_color(color::pink);
    if (c == 3)
        set_pen_color(color::purple);
    if (c == 4)
        set_pen_color(color::light_red);
    if (c == 5)
        set_pen_color(color::indigo);
    if (c == 6)
        set_pen_color(color::light_green);
    if (c == 7)
        set_pen_color(color::green);
    if (c == 8)
        set_pen_color(color::yellow);
    if (c == 9)
        set_pen_color(color::violet);
    if (c == 10)
        set_pen_color(color::red);
    if (c == 11)
        set_pen_color(color::light_grey);
    if (c == 12)
        set_pen_color(color::white);
    if (c == 13)
        set_pen_color(color::grey);
    if (c == 14)
        set_pen_color(color::dark_blue);
}

```

```

while (number_of_windows < n)
{

    current_x = current_x + w + space_x;
    window(current_x, y, w, h);

    number_of_windows = number_of_windows + 1;
}
}

```

```

void block_of_windows(const int x, const int y, const int w, const int h, const int n_floors, const
int n_windows, double space_x, double space_y, const int c)

```

```

{
    int number_of_floors = 0;
    int sx = space_x;
    int sy = space_y;
    int current_y = y;

    if (c == 1)
        set_pen_color(color::orange);
    if (c == 2)
        set_pen_color(color::pink);
    if (c == 3)
        set_pen_color(color::purple);
    if (c == 4)
        set_pen_color(color::light_red);
    if (c == 5)
        set_pen_color(color::indigo);
    if (c == 6)
        set_pen_color(color::light_green);
    if (c == 7)
        set_pen_color(color::green);
    if (c == 8)
        set_pen_color(color::yellow);
    if (c == 9)
        set_pen_color(color::violet);
    if (c == 10)
        set_pen_color(color::red);
    if (c == 11)
        set_pen_color(color::light_grey);
    if (c == 12)
        set_pen_color(color::white);
    if (c == 13)

```

```

        set_pen_color(color::grey);
    if (c == 14)
        set_pen_color(color::dark_blue);

    while (number_of_floors < n_floors)
    {
        row_of_windows(x, current_y, w, h, n_windows, space_x, c);
        current_y = current_y + h + space_y;

        number_of_floors = number_of_floors + 1;
    }
}

void office_building(const int x, const int ground_y, int w)
{
    const int n_floors = random_in_range(4, 10);
    int window_width = 30;
    int window_height = 35;
    int space_x = 8;
    int space_y = 8;
    int top_bottom_margin = 10;

    int building_height = n_floors * (window_height + space_y) + top_bottom_margin * 2;

    //STAY ON THE GROUND!!!!
    int y = 570 - building_height;

    int building_color = random_in_range(1, 14);
    fill_rectangle(x, y, w, building_height);
    int window_color = random_in_range(1, 14);

    while (window_color == building_color)
    {
        window_color = random_in_range(1, 14);
    }

    int available_width = w - space_y;

```

```

int windows_per_floor = available_width / (window_width + space_y);

int window_start_y = y + top_bottom_margin;

block_of_windows(x - 27.5, window_start_y, window_width, window_height, n_floors,
windows_per_floor, space_x, space_y, window_color);
}

```

```

void fifty_stars(int x, int y)
{
    set_pen_color(color::white);

```

```

    fill_rectangle(x + 2, y + 5, 1, 1);
    fill_rectangle(x + 4, y + 5, 1, 1);
    fill_rectangle(x + 6, y + 5, 1, 1);
    fill_rectangle(x + 8, y + 5, 1, 1);
    fill_rectangle(x + 10, y + 5, 1, 1);
    fill_rectangle(x + 12, y + 5, 1, 1);

```

```

    fill_rectangle(x + 3, y + 6, 1, 1);
    fill_rectangle(x + 5, y + 6, 1, 1);
    fill_rectangle(x + 7, y + 6, 1, 1);
    fill_rectangle(x + 9, y + 6, 1, 1);
    fill_rectangle(x + 11, y + 6, 1, 1);

```

```

    fill_rectangle(x + 2, y + 7, 1, 1);
    fill_rectangle(x + 4, y + 7, 1, 1);
    fill_rectangle(x + 6, y + 7, 1, 1);
    fill_rectangle(x + 8, y + 7, 1, 1);
    fill_rectangle(x + 10, y + 7, 1, 1);
    fill_rectangle(x + 12, y + 7, 1, 1);

```

```

    fill_rectangle(x + 3, y + 8, 1, 1);
    fill_rectangle(x + 5, y + 8, 1, 1);
    fill_rectangle(x + 7, y + 8, 1, 1);
    fill_rectangle(x + 9, y + 8, 1, 1);
    fill_rectangle(x + 11, y + 8, 1, 1);

```

```

    fill_rectangle(x + 2, y + 9, 1, 1);
    fill_rectangle(x + 4, y + 9, 1, 1);
    fill_rectangle(x + 6, y + 9, 1, 1);
    fill_rectangle(x + 8, y + 9, 1, 1);

```

```

    fill_rectangle(x + 10, y + 9, 1, 1);
    fill_rectangle(x + 12, y + 9, 1, 1);
}

```

```

void mini_american_flag(int x, int y)
{
    set_pen_color(color::white);
    fill_rectangle(x, y, 25, 13);
    set_pen_color(color::red);
    fill_rectangle(x, y, 25, 2);
    fill_rectangle(x, y + 4, 25, 2);
    fill_rectangle(x, y + 8, 25, 2);
        fill_rectangle(x, y + 13, 25, 2);

    set_pen_color(color::blue);
    fill_rectangle(x, y, 10, 5);

        fifty_stars(x-2, y-5);
}

```

```

void variety(const int x, const int y, const int w)
{
    int choice = random_in_range(1, 2);

    if (choice == 1) //chimney
    {
        set_pen_color(color::dark_grey);
        fill_rectangle(x + w - 30, y - 50, 10, 40);
    }

    else if (choice == 2) //flag
    {
        set_pen_color(color::dark_grey);
        fill_rectangle(x + w/2 + 40, y - 60, 3, 60);
            move_to(x + w/2 + 40, y - 60);
            mini_american_flag(x + w/2 + 40, y - 60);

    }

}

```

```

void house(const int x, const int ground_y, int w)
{
    const int n_floors = random_in_range(1, 2);

    int window_width = 30;
    int window_height = 35;
    int space_x = 30;
    int space_y = 8;
    int top_bottom_margin = 10;

    int house_height = n_floors * (window_height + space_y) + top_bottom_margin * 2;
    int y = ground_y - house_height;

    int house_color = random_in_range(1, 8);
    fill_rectangle(x, y, w, house_height);
    variety(x,y,w);

    start_shape();
    move_to(x, y);
    note_position();
    move_to(x + w / 2, y - 40);
    note_position();
    move_to(x + w, y);
    note_position();
    set_pen_color(color::brown);
    fill_shape();

    int available_width = w - space_y;
    int windows_per_floor = available_width / (window_width + space_y);
    int window_color = random_in_range(1, 14);

    while (window_color == house_color)
    {
        window_color = random_in_range(1, 14);
    }

    int window_start_y = y + top_bottom_margin;
    block_of_windows(x-55, window_start_y, window_width, window_height, n_floors,
    windows_per_floor, space_x, space_y, window_color);
}

```

```
int door_color = random_in_range(1,14);
int door_x = x + 35;
int door_y = (ground_y - window_height - 5);
fill_rectangle(door_x, door_y, window_width, window_height + 10);

}
```

```
void city_scene()
{
    int x = 20;

    while (x < 800)
    {
        int ground_y = 240;
        int width = random_in_range(120, 180);
        office_building(x, ground_y, width);
        x = x + width + 15;
    }

    x = 20;

    while (x < 800)
    {
        int ground_y = 570;
        int width = 100;
        house(x, ground_y, width);
        x = x + width + 70;
    }
}
```

```
void main()
{
    background();
    city_scene();
}
```



```
#include "library.h"
```

```
void circle_steps(double step_length, double turn_angle, int steps_left)
{
    if (steps_left == 0)
    {
        return;
    }
    note_position();
    draw_distance(step_length);
    turn_right_by_degrees(turn_angle);
    circle_steps(step_length, turn_angle, steps_left - 1);
}
```

```
void sun_or_moon(double x, double y, double radius, const int steps, int choose_sky)
{
    double pi = acos(-1.0);
    double circumference = 2 * pi * radius;
    double step_length = circumference / steps;
    double turn_angle = 360 / steps;

    start_shape();
    move_to(x + radius, y);
    set_heading_degrees(90);

    circle_steps(step_length, turn_angle, steps);
    note_position();

    if (choose_sky == 1)
        set_pen_color(color::yellow);
    else set_pen_color(color::light_grey);

    fill_shape();
}
```

```
void grass()
{
    set_pen_color(color::light_green);
    fill_rectangle(0,550,800,570);
}
```

```
}
```

```
void dirt()
```

```
{  
    set_pen_color(color::brown);  
    fill_rectangle(0,570,800,600);  
}
```

```
void background()
```

```
{  
    make_window(800, 600);  
  
    int choose_sky = random_in_range(1, 2);  
  
    //background  
    if (choose_sky == 1)  
        set_pen_color(color::light_blue);  
    else  
        set_pen_color(color::dark_blue);  
    fill_rectangle(0, 0, 800, 550);  
  
    //ground  
    grass();  
    dirt();  
  
    //sun or moon or wateva  
    if (choose_sky == 1)  
        sun_or_moon(600, 70, 30, 360, 1);  
    else  
        sun_or_moon(600, 70, 30, 360, 2);  
}
```

```
void building(const int x, const int y, const int w, const int h)
```

```
{  
    const int width = x + w;  
    const int height = y + h;  
  
    set_pen_color(color::pink);  
    fill_rectangle(x, y, w, h);  
}
```

```

void window(const int x, const int y, const int w, const int h)
{
    const int width = x + w;
    const int height = y + h;

    fill_rectangle(x, y, w, h);
}

```

```

void row_of_windows(const int x, const int y, const int w, const int h, const int n, double
space_x, const int c)
{
    int number_of_windows = 0;
    int current_x = x;

    if (c == 1)
        set_pen_color(color::orange);
    if (c == 2)
        set_pen_color(color::pink);
    if (c == 3)
        set_pen_color(color::purple);
    if (c == 4)
        set_pen_color(color::light_red);
    if (c == 5)
        set_pen_color(color::indigo);
    if (c == 6)
        set_pen_color(color::light_green);
    if (c == 7)
        set_pen_color(color::green);
    if (c == 8)
        set_pen_color(color::yellow);
    if (c == 9)
        set_pen_color(color::violet);
    if (c == 10)
        set_pen_color(color::red);
    if (c == 11)
        set_pen_color(color::light_grey);
    if (c == 12)
        set_pen_color(color::white);
    if (c == 13)
        set_pen_color(color::grey);
    if (c == 14)
        set_pen_color(color::dark_blue);
}

```

```

while (number_of_windows < n)
{

    current_x = current_x + w + space_x;
    window(current_x, y, w, h);

    number_of_windows = number_of_windows + 1;

}
}

```

```

void block_of_windows(const int x, const int y, const int w, const int h, const int n_floors, const
int n_windows, double space_x, double space_y, const int c)

```

```

{
    int number_of_floors = 0;
    int sx = space_x;
    int sy = space_y;
    int current_y = y;

    if (c == 1)
        set_pen_color(color::orange);
    if (c == 2)
        set_pen_color(color::pink);
    if (c == 3)
        set_pen_color(color::purple);
    if (c == 4)
        set_pen_color(color::light_red);
    if (c == 5)
        set_pen_color(color::indigo);
    if (c == 6)
        set_pen_color(color::light_green);
    if (c == 7)
        set_pen_color(color::green);
    if (c == 8)
        set_pen_color(color::yellow);
    if (c == 9)
        set_pen_color(color::violet);
    if (c == 10)
        set_pen_color(color::red);
    if (c == 11)
        set_pen_color(color::light_grey);
}

```

```

    if (c == 12)
        set_pen_color(color::white);
    if (c == 13)
        set_pen_color(color::grey);
    if (c == 14)
        set_pen_color(color::dark_blue);

    while (number_of_floors < n_floors)
    {
        row_of_windows(x, current_y, w, h, n_windows, space_x, c);
        current_y = current_y + h + space_y;

        number_of_floors = number_of_floors + 1;
    }
}

void office_building(const int x, const int ground_y, int w)
{
    const int n_floors = random_in_range(4, 10);
    int window_width = 30;
    int window_height = 35;
    int space_x = 8;
    int space_y = 8;
    int top_bottom_margin = 10;

    int building_height = n_floors * (window_height + space_y) + top_bottom_margin * 2;

    //STAY ON THE GROUND!!!!
    int y = 570 - building_height;

    int building_color = random_in_range(1, 14);
    fill_rectangle(x, y, w, building_height);
    int window_color = random_in_range(1, 14);

    while (window_color == building_color)
    {
        window_color = random_in_range(1, 14);
    }
}

```

```

    }

    int available_width = w - space_y;
    int windows_per_floor = available_width / (window_width + space_y);

    int window_start_y = y + top_bottom_margin;

    block_of_windows(x - 27.5, window_start_y, window_width, window_height, n_floors,
    windows_per_floor, space_x, space_y, window_color);
}

```

```

void fifty_stars(int x, int y)
{
    set_pen_color(color::white);

```

```

    fill_rectangle(x + 2, y + 5, 1, 1);
    fill_rectangle(x + 4, y + 5, 1, 1);
    fill_rectangle(x + 6, y + 5, 1, 1);
    fill_rectangle(x + 8, y + 5, 1, 1);
    fill_rectangle(x + 10, y + 5, 1, 1);
    fill_rectangle(x + 12, y + 5, 1, 1);

```

```

    fill_rectangle(x + 3, y + 6, 1, 1);
    fill_rectangle(x + 5, y + 6, 1, 1);
    fill_rectangle(x + 7, y + 6, 1, 1);
    fill_rectangle(x + 9, y + 6, 1, 1);
    fill_rectangle(x + 11, y + 6, 1, 1);

```

```

    fill_rectangle(x + 2, y + 7, 1, 1);
    fill_rectangle(x + 4, y + 7, 1, 1);
    fill_rectangle(x + 6, y + 7, 1, 1);
    fill_rectangle(x + 8, y + 7, 1, 1);
    fill_rectangle(x + 10, y + 7, 1, 1);
    fill_rectangle(x + 12, y + 7, 1, 1);

```

```

    fill_rectangle(x + 3, y + 8, 1, 1);
    fill_rectangle(x + 5, y + 8, 1, 1);
    fill_rectangle(x + 7, y + 8, 1, 1);
    fill_rectangle(x + 9, y + 8, 1, 1);
    fill_rectangle(x + 11, y + 8, 1, 1);

```

```

    fill_rectangle(x + 2, y + 9, 1, 1);

```

```

    fill_rectangle(x + 4, y + 9, 1, 1);
    fill_rectangle(x + 6, y + 9, 1, 1);
    fill_rectangle(x + 8, y + 9, 1, 1);
    fill_rectangle(x + 10, y + 9, 1, 1);
    fill_rectangle(x + 12, y + 9, 1, 1);
}

```

```

void mini_american_flag(int x, int y)
{
    set_pen_color(color::white);
    fill_rectangle(x, y, 25, 13);
    set_pen_color(color::red);
    fill_rectangle(x, y, 25, 2);
    fill_rectangle(x, y + 4, 25, 2);
    fill_rectangle(x, y + 8, 25, 2);
    fill_rectangle(x, y + 13, 25, 2);

    set_pen_color(color::blue);
    fill_rectangle(x, y, 10, 5);

    fifty_stars(x-2, y-5);
}

```

```

void variety(const int x, const int y, const int w)
{
    int choice = random_in_range(1, 2);

    if (choice == 1) //chimney
    {
        set_pen_color(color::dark_grey);
        fill_rectangle(x + w - 30, y - 50, 10, 40);
    }

    else if (choice == 2) //flag
    {
        set_pen_color(color::dark_grey);
        fill_rectangle(x + w/2 + 40, y - 60, 3, 60);
        move_to(x + w/2 + 40, y - 60);
        mini_american_flag(x + w/2 + 40, y - 60);
    }
}

```

```
}
```

```
void house(const int x, const int ground_y, int w)
```

```
{
```

```
    const int n_floors = random_in_range(1, 2);
```

```
    int window_width = 30;
```

```
    int window_height = 35;
```

```
        int space_x = 30;
```

```
    int space_y = 8;
```

```
    int top_bottom_margin = 10;
```

```
    int house_height = n_floors * (window_height + space_y) + top_bottom_margin * 2;
```

```
    int y = ground_y - house_height;
```

```
    int house_color = random_in_range(1, 8);
```

```
    fill_rectangle(x, y, w, house_height);
```

```
        variety(x,y,w);
```

```
    start_shape();
```

```
    move_to(x, y);
```

```
    note_position();
```

```
    move_to(x + w / 2, y - 40);
```

```
    note_position();
```

```
    move_to(x + w, y);
```

```
    note_position();
```

```
    set_pen_color(color::brown);
```

```
    fill_shape();
```

```
    int available_width = w - space_y;
```

```
    int windows_per_floor = available_width / (window_width + space_y);
```

```
    int window_color = random_in_range(1, 14);
```

```
        while (window_color == house_color)
```

```
        {
```

```
            window_color = random_in_range(1, 14);
```

```
        }
```

```
    int window_start_y = y + top_bottom_margin;
```

```
    block_of_windows(x-55, window_start_y, window_width, window_height, n_floors,  
windows_per_floor, space_x, space_y, window_color);
```

```
    int door_color = random_in_range(1,14);  
    int door_x = x + 35;  
    int door_y = (ground_y - window_height - 5);  
    fill_rectangle(door_x, door_y, window_width, window_height + 10);  
  
}
```

```
void circle(double x, double y, double radius, const int steps)  
{  
    double pi = acos(-1.0);  
    double circumference = 2 * pi * radius;  
    double step_length = circumference / steps;  
    double turn_angle = 360 / steps;  
  
    start_shape();  
    move_to(x + radius, y);  
    set_heading_degrees(90);  
    circle_steps(step_length, turn_angle, steps);  
    note_position();  
  
    int color = random_in_range(1,3);  
    if (color == 1)  
        set_pen_color(color::light_green);  
    if (color == 2)  
        set_pen_color(color::green);  
    else set_pen_color(color::dark_green);  
  
    fill_shape();  
}
```

```
void trees(int x, int ground_y)  
{  
    int height = random_in_range(50, 150);  
    int width = random_in_range(12, 18);  
  
    set_pen_color(color::brown);  
    fill_rectangle(x, ground_y - height, width, height);  
}
```

```

int leaves = random_in_range(100,600);

int i = 0;
while (i < leaves)
{
    circle(x + width/2 + random_in_range(-30, 30), ground_y - height - random_in_range(0,
70), 3, 360);
    i = i + 1;
}
}

void city_scene()
{
    int x = 20;

    while (x < 800)
    {
        int ground_y = 240;
        int width = random_in_range(120, 180);
        office_building(x, ground_y, width);
        x = x + width + 15;
    }

    x = 20;

    while (x < 800)
    {
        int ground_y = 570;
        int width = 100;
        house(x, ground_y, width);
        x = x + width + 70;
    }

    x = 100;

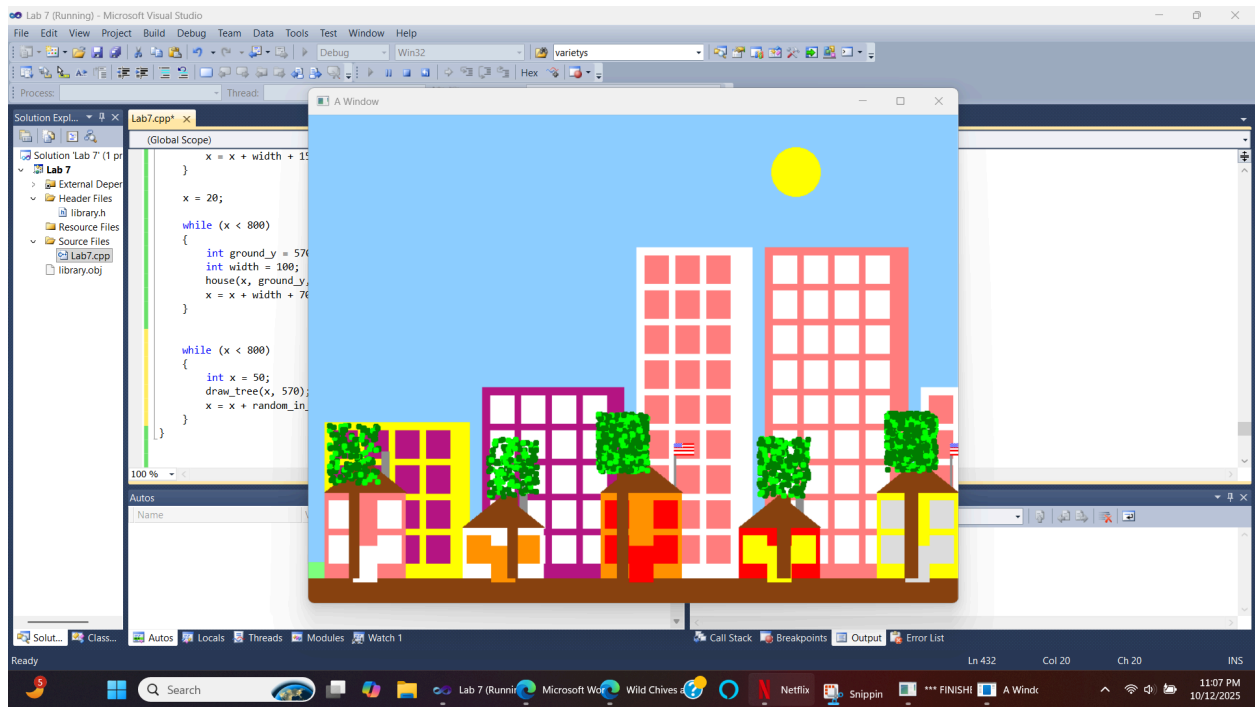
    while (x < 800)
    {
        trees(x, 570);
        x = x + random_in_range(120, 200);
    }
}

```

```

void main()
{
    background();
    city_scene();
}

```



6. Make it look nice

```
#include "library.h"
```

```
void circle_steps(double step_length, double turn_angle, int steps_left)
{
    if (steps_left == 0)
    {
        return;
    }
    note_position();
    draw_distance(step_length);
    turn_right_by_degrees(turn_angle);
    circle_steps(step_length, turn_angle, steps_left - 1);
}
```

```
void sun_or_moon(double x, double y, double radius, const int steps, int choose_sky)
{
    double pi = acos(-1.0);
    double circumference = 2 * pi * radius;
    double step_length = circumference / steps;
    double turn_angle = 360 / steps;

    start_shape();

    move_to(x + radius, y);
    set_heading_degrees(90);
    circle_steps(step_length, turn_angle, steps);
    note_position();

    if (choose_sky == 1)
        set_pen_color(color::yellow);
    else
        set_pen_color(color::light_grey);

    fill_shape();
}
```

```
void grass()
{
    set_pen_color(color::light_green);
```

```
    fill_rectangle(0, 550, 800, 570);  
}
```

```
void dirt()  
{  
    set_pen_color(color::brown);  
    fill_rectangle(0, 570, 800, 600);  
}
```

```
void background()  
{  
    make_window(800, 600);  
  
    int choose_sky = random_in_range(1, 2);  
  
    if (choose_sky == 1)  
        set_pen_color(color::light_blue);  
    else  
        set_pen_color(color::dark_blue);  
  
    fill_rectangle(0, 0, 800, 550);  
  
    grass();  
    dirt();  
  
    if (choose_sky == 1)  
        sun_or_moon(600, 70, 30, 360, 1);  
    else  
        sun_or_moon(600, 70, 30, 360, 2);  
}
```

```
void plant_bush(double x, double y, double radius, const int steps)  
{  
    double pi = acos(-1.0);  
    double circumference = 2 * pi * radius;  
    double step_length = circumference / steps;  
    double turn_angle = 360 / steps;  
  
    start_shape();  
  
    move_to(x + radius, y);
```

```
    set_heading_degrees(90);
    circle_steps(step_length, turn_angle, steps);
    note_position();

    fill_shape();
}
```

```
void random_color(int c)
{
    if (c == 1)
        set_pen_color(color::orange);
    if (c == 2)
        set_pen_color(color::pink);
    if (c == 3)
        set_pen_color(color::purple);
    if (c == 4)
        set_pen_color(color::light_red);
    if (c == 5)
        set_pen_color(color::indigo);
    if (c == 6)
        set_pen_color(color::light_green);
    if (c == 7)
        set_pen_color(color::green);
    if (c == 8)
        set_pen_color(color::yellow);
    if (c == 9)
        set_pen_color(color::violet);
    if (c == 10)
        set_pen_color(color::red);
    if (c == 11)
        set_pen_color(color::light_grey);
    if (c == 12)
        set_pen_color(color::white);
    if (c == 13)
        set_pen_color(color::grey);
    if (c == 14)
        set_pen_color(color::dark_blue);
}
```

```
void window_decorations(int x, int y, int w, int h)
{
```

```

int choice = random_in_range(1, 3);

if (choice == 1) //curtains
{
    set_pen_color(color::light_red);
    fill_rectangle(x + 3, y + 3, w / 2 - 4, h - 6);
    fill_rectangle(x + w / 2 + 1, y + 3, w / 2 - 4, h - 6);
}
else if (choice == 2) //cat
{
    int cat_color = random_in_range(11,14);
    set_pen_color(cat_color);
    fill_rectangle(x + w / 2 - 2, y + h - 8, 5, 5);
    fill_rectangle(x + w / 2 - 1, y + h - 12, 3, 3);
    fill_rectangle(x + w / 2 - 2, y + h - 13, 1, 1);
    fill_rectangle(x + w / 2 + 2, y + h - 13, 1, 1);
}
else if (choice == 3) //house plant
{
    set_pen_color(color::dark_green);
    fill_rectangle(x + w / 2 - 1, y + h - 8, 3, 6);
    set_pen_color(color::green);
    plant_bush(x + w / 2 - 3, y + h - 15, 3, 360);
}
}

```

```

void window(const int x, const int y, const int w, const int h)
{
    fill_rectangle(x, y, w, h);
    window_decorations(x, y, w, h);
}

```

```

void row_of_windows(const int x, const int y, const int w, const int h, const int n, double
space_x)
{
    int number_of_windows = 0;
    int current_x = x;

    while (number_of_windows < n)
    {
        current_x = current_x + w + space_x;
        window(current_x, y, w, h);
    }
}

```

```

        number_of_windows = number_of_windows + 1;
    }
}

```

```

void block_of_windows(const int x, const int y, const int w, const int h, const int n_floors, const
int n_windows, double space_x, double space_y, const int c)
{
    int number_of_floors = 0;
    int current_y = y;

    random_color(c);

    while (number_of_floors < n_floors)
    {
        row_of_windows(x, current_y, w, h, n_windows, space_x);
        current_y = current_y + h + space_y;
        number_of_floors = number_of_floors + 1;
    }
}

```

```

void office_building(const int x, const int ground_y, int w)
{
    const int n_floors = random_in_range(4, 10);
    int window_width = 30;
    int window_height = 35;
    int space_x = 8;
    int space_y = 8;
    int top_bottom_margin = 10;

    int building_height = n_floors * (window_height + space_y) + top_bottom_margin * 2;
    int y = 570 - building_height;

    int building_color = random_in_range(1, 10);
    random_color(building_color);
    fill_rectangle(x, y, w, building_height);

    int window_color = random_in_range(1, 14);
    while (window_color == building_color)
    {
        window_color = random_in_range(1, 14);
    }
}

```

```

int available_width = w - space_y;
int windows_per_floor = available_width / (window_width + space_y);

int window_start_y = y + top_bottom_margin;
block_of_windows(x - 27.5, window_start_y, window_width, window_height, n_floors,
windows_per_floor, space_x, space_y, window_color);
}

```

```

void fifty_stars(int x, int y)
{
    set_pen_color(color::white);

```

```

    fill_rectangle(x + 2, y + 5, 1, 1);
    fill_rectangle(x + 4, y + 5, 1, 1);
    fill_rectangle(x + 6, y + 5, 1, 1);
    fill_rectangle(x + 8, y + 5, 1, 1);
    fill_rectangle(x + 10, y + 5, 1, 1);
    fill_rectangle(x + 12, y + 5, 1, 1);

```

```

    fill_rectangle(x + 3, y + 6, 1, 1);
    fill_rectangle(x + 5, y + 6, 1, 1);
    fill_rectangle(x + 7, y + 6, 1, 1);
    fill_rectangle(x + 9, y + 6, 1, 1);
    fill_rectangle(x + 11, y + 6, 1, 1);

```

```

    fill_rectangle(x + 2, y + 7, 1, 1);
    fill_rectangle(x + 4, y + 7, 1, 1);
    fill_rectangle(x + 6, y + 7, 1, 1);
    fill_rectangle(x + 8, y + 7, 1, 1);
    fill_rectangle(x + 10, y + 7, 1, 1);
    fill_rectangle(x + 12, y + 7, 1, 1);

```

```

    fill_rectangle(x + 3, y + 8, 1, 1);
    fill_rectangle(x + 5, y + 8, 1, 1);
    fill_rectangle(x + 7, y + 8, 1, 1);
    fill_rectangle(x + 9, y + 8, 1, 1);
    fill_rectangle(x + 11, y + 8, 1, 1);

```

```

    fill_rectangle(x + 2, y + 9, 1, 1);
    fill_rectangle(x + 4, y + 9, 1, 1);
    fill_rectangle(x + 6, y + 9, 1, 1);
    fill_rectangle(x + 8, y + 9, 1, 1);

```

```
    fill_rectangle(x + 10, y + 9, 1, 1);
    fill_rectangle(x + 12, y + 9, 1, 1);
}
```

```
void mini_american_flag(int x, int y)
{
    set_pen_color(color::white);
    fill_rectangle(x, y, 25, 13);
    set_pen_color(color::red);
    fill_rectangle(x, y, 25, 2);
    fill_rectangle(x, y + 4, 25, 2);
    fill_rectangle(x, y + 8, 25, 2);

    set_pen_color(color::blue);
    fill_rectangle(x, y, 10, 5);
    fifty_stars(x - 2, y - 5);
}
```

```
void variety(const int x, const int y, const int w)
{
    int choice = random_in_range(1, 2);

    if (choice == 1) // chimney
    {
        set_pen_color(color::dark_grey);
        fill_rectangle(x + w - 30, y - 50, 10, 40);
    }

    else if (choice == 2) // flag
    {
        set_pen_color(color::dark_grey);
        fill_rectangle(x + w / 2 + 40, y - 60, 3, 60);
        mini_american_flag(x + w / 2 + 40, y - 60);
    }
}
```

```
void house(const int x, const int ground_y, int w)
{
    const int n_floors = random_in_range(1, 2);
```

```

int window_width = 30;
int window_height = 35;
int space_x = 30;
int space_y = 8;
int top_bottom_margin = 10;

int house_height = n_floors * (window_height + space_y) + top_bottom_margin * 2;
int y = ground_y - house_height;

int house_color = random_in_range(6, 10);
random_color(house_color);
fill_rectangle(x, y, w, house_height);
variety(x, y, w);

// roof
start_shape();
move_to(x, y);
note_position();
move_to(x + w/2, y - 40);
note_position();
move_to(x + w, y);
note_position();
set_pen_color(color::brown);
fill_shape();

int available_width = w - space_y;
int windows_per_floor = available_width / (window_width + space_y);
int window_color = random_in_range(1, 14);

while (window_color == house_color)
{
    window_color = random_in_range(1, 14);
}

int window_start_y = y + top_bottom_margin;
block_of_windows(x - 55, window_start_y, window_width, window_height, n_floors,
windows_per_floor, space_x, space_y, window_color);

int door_color = random_in_range(1, 14);
random_color(door_color);
int door_x = x + 35;
int door_y = ground_y - window_height - 5;
fill_rectangle(door_x, door_y, window_width, window_height + 10);
}

```

```

void circle_leaves(double x, double y, double radius, const int steps)
{
    double pi = acos(-1.0);
    double circumference = 2 * pi * radius;
    double step_length = circumference / steps;
    double turn_angle = 360 / steps;

    start_shape();

    move_to(x + radius, y);
    set_heading_degrees(90);
    circle_steps(step_length, turn_angle, steps);
    note_position();

    int color = random_in_range(1,3);
    if (color == 1)
        set_pen_color(color::light_green);
    if (color == 2)
        set_pen_color(color::green);
    else set_pen_color(color::dark_green);

    fill_shape();
}

```

```

void trees(int x, int ground_y)
{
    int height = random_in_range(50, 150);
    int width = random_in_range(12, 18);

    set_pen_color(color::brown);
    fill_rectangle(x, ground_y - height, width, height);

    int leaves = random_in_range(100, 300);
    int i = 0;
    while (i < leaves)
    {
        set_pen_color(color::green);
        circle_leaves(x + width / 2 + random_in_range(-30, 30), ground_y - height -
random_in_range(0, 70), 3, 360);
        i = i + 1;
    }
}

```

```
}  
}
```

```
void circle_cloud(double x, double y, double radius, const int steps)
```

```
{  
    double pi = acos(-1.0);  
    double circumference = 2 * pi * radius;  
    double step_length = circumference / steps;  
    double turn_angle = 360 / steps;  
  
    start_shape();  
  
    move_to(x + radius, y);  
    set_heading_degrees(90);  
    circle_steps(step_length, turn_angle, steps);  
    note_position();  
  
    set_pen_color(color::white);  
  
    fill_shape();  
}
```

```
void cloud(int x, int y)
```

```
{  
    set_pen_color(color::white);  
    circle_cloud(x, y, 15, 360);  
    circle_cloud(x + 20, y, 20, 360);  
    circle_cloud(x + 40, y, 15, 360);  
}
```

```
void clouds()
```

```
{  
    int i = 0;  
  
    while (i < 6)  
    {  
        int cloud_x = random_in_range(50, 750);  
        int cloud_y = random_in_range(40, 200);  
        cloud(cloud_x, cloud_y);  
        i = i + 1;  
    }  
}
```

```

void city_scene()
{
    int x = 20;
    while (x < 800)
    {
        int ground_y = 240;
        int width = random_in_range(120, 180);
        office_building(x, ground_y, width);
        x = x + width + 15;
    }

    x = 20;
    while (x < 800)
    {
        int ground_y = 570;
        int width = 100;
        house(x, ground_y, width);
        x = x + width + 70;
    }

    x = 100;
    while (x < 800)
    {
        trees(x, 570);
        x = x + random_in_range(120, 200);
    }

    clouds();
}

```

```

void main()
{
    background();
    city_scene();
}

```

